

LOST IN TRANSMISSION

INDIA'S SOLAR CAPACITY HAS INCREASED, BUT THE LACK OF AN EFFICIENT EVACUATION INFRASTRUCTURE IS IMPACTING ITS ADOPTION

RICHA SHARMA

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WILL INDIA MISS its 500 GW renewable electricity target by 2030? A growing mismatch between

buildout of renewable projects and grid infrastructure is emerging as the most critical risk to the country's green energy goals.

Project developers Tata Power, Adani, JSW, and NTPC Green were forced to reduce solar power production last fiscal in Maharashtra, Rajasthan, and Gujarat, the states with the highest solar power output, due to congested grid. This, in fact, impacted Tata Power's earnings as well. Project developers are also delaying the addition of new solar power capacity in the absence of adequate grid infrastructure, say industry watchers.

India lost around 34 gigawatt-

hours (GWh) of renewable energy generation in a single day on March 30 due to insufficient transmission margins, equivalent to the daily power consumption of about five million middle-class, urban households in India, according to Ember, a global energy think tank. In the first quarter of this year, transmission constraints accounted for nearly two-thirds of all renewable energy curtailment at 300 GWh.

The targets for building the evacuation infrastructure are now significantly higher than historical build rates. There is a reason for that. Transmission planning timelines have traditionally synchronised with thermal capacity, which typically takes 36–60 months to come online. In contrast, renewable energy

